

A fuzzy ubiquitous traveler clustering and hotel recommendation system by differentiating travelers' decision-making behaviors

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ARTICLE INFO

Article history:

Received 12 August 2018

Received in revised form 29 May 2019

Accepted 24 July 2020

Available online 28 July 2020

Keywords:

Ubiquitous recommendation
Fuzzy mixed binary-nonlinear
programming
Clustering

ABSTRACT

For generating hotel recommendations, clustering travelers has been demonstrated to be a viable method to elevate traveler satisfaction with the recommendation results. However, most of the existing methods that adopt this approach cluster travelers according to a variety of traveler or hotel attributes, which may not necessarily be appropriate for use in an online application such as ubiquitous hotel recommendation. To overcome this problem, a fuzzy ubiquitous traveler clustering and hotel recommendation (FUTCHR) system was developed in this study. The FUTCHR system clustered travelers according to their decision-making mechanisms that are fitted by comparing travelers' choices with the recommendation results in the historical data. To generate recommendations, a fuzzy mixed binary-nonlinear programming model was constructed and solved. The novelty of the proposed methodology is to cluster travelers without knowing their characteristics but according to the differences in their decision-making mechanisms. The FUTCHR system was employed in a regional study, and the successful recommendation rate was superior to three existing methods in this field.

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1. Introduction

Advances in communication and computation technologies have inspired many innovative ideas on location-based services and recommendation systems [1–4]. The intersection of the two fields includes a variety of ubiquitous recommendation systems that recommend the most suitable service, facility, or resource to a mobile user on the basis of the user's location, preferences, and other context-related information [5]: for example, ubiquitous clinic recommendation systems [6], ubiquitous tourist attraction recommendation systems [7], ubiquitous restaurant recommendation systems [8], and ubiquitous hotel recommendation systems [9]. Ubiquitous recommendation systems are a type of location-based services, an important topic in ambient intelligence, to which artificial intelligence and soft computing have been extensively applied [10–12]. Among these, ubiquitous hotel recommendation systems have received considerable attention because a traveler will typically select a hotel before searching for nearby tourist attractions or restaurants [13,14].

This study aims to improve the effectiveness of a ubiquitous hotel recommendation system, in terms of the successful recommendation rate (i.e., the percentage of travelers that followed the recommendations), under the condition that a traveler is not willing or convenient to articulate his/her preferences. The existing ubiquitous hotel recommendation mechanisms present two problems:

- (1) The cold-start (or sparsity) problem is the situation in which the decision-making behaviors of a few travelers are markedly different from the majority. This can be addressed by recommending multiple hotels that are as diverse as possible to a traveler [15]. Clustering travelers is another possible method to overcome the cold-start problem [16].
- (2) The scalability problem is where the time required for making recommendations to a number of travelers in parallel increases drastically as the number of travelers increases [17]. Forming clusters of travelers and designing a recommendation mechanism for each cluster can resolve the scalability problem [16].

Traveler clustering presents a viable means to overcome these two key problems. For this reason, a ubiquitous hotel recommendation system that clusters travelers by differentiating their decision-making behaviors using fuzzy logic is developed in this study. The system is called the fuzzy ubiquitous traveler clustering and hotel recommendation (FUTCHR) system. The FUTCHR system detects the location of a traveler, but does not require additional inputs from the traveler to make a recommendation. Travelers are divided into multiple clusters in which the travelers within a cluster exhibit similar decision-making behaviors. As a consequence, the FUTCHR system applies various mechanisms to recommend suitable hotels to travelers in different clusters. The proposed methodology assumes that travelers assess the overall performance of a hotel by using a mechanism similar

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to the fuzzy weighted average (FWA) mechanism. Therefore, for each traveler cluster an individual FWA mechanism must be fitted. This is accomplished through a comparison of the recommended hotels and travelers' choices evident in the historical data. To determine the fuzzy weights in the FWA mechanisms, a fuzzy mixed binary-nonlinear programming (FMBNLP) model is proposed. To facilitate problem solving, the FMBNLP model is converted into a mixed binary-nonlinear programming (MBNLP) problem to be solved. The novelty of the proposed methodology is to cluster travelers without knowing their characteristics but according to the differences in their decision-making mechanisms. To ascertain whether the FUTCHR system achieves a better successful recommendation rate than the existing systems, a regional experiment is conducted. In the regional experiment, the FUTCHR system recommends the suitable hotel to each traveler, and records his/her choice, so as to evaluate the successful recommendation rate. The collected data are also analyzed using the existing recommendation mechanisms to compare the successful recommendation rates achieved by various systems.

The remaining sections of this paper are organized as follows. Section 2 reviews some of the relevant literature and the practice of ubiquitous hotel recommendation. Section 3 introduces the FUTCHR system and its operational procedure, the FMBNLP model, and the MBNLP problem. A numerical example is also provided to illustrate the applicability of the FUTCHR system. To assess the effectiveness of the FUTCHR system, a regional study was conducted in Hsinchu City, Taiwan. Some of the existing methods were also applied in the regional study for comparison. Finally, Section 4 concludes the findings and suggests some directions for future investigation.

2. Literature review

In practice, the smartphone is the most prevalent user interface for ubiquitous hotel recommendations. Although a smartphone can support sophisticated operations, it is still not comparable to a personal computer in its capacity for browsing or processing a large quantity of data or information. The conditions or environment in which a smartphone is used can also cause some inconvenience.

In theory, two extremes of ubiquitous hotel recommendation methods exist. One method does not need inputs from a traveler, but the other relies on massive inputs. Prevalent ubiquitous hotel recommendation systems such as Google Maps, TripAdvisor, and many navigation applications (apps) recommend the nearest or cheapest hotel to a traveler only on the basis of the traveler's location. The other method permits travelers to input their preferences such room price, hotel rating [18,19], and hotel class to obtain a more precise recommendation result. Other context-related information include the number of guests, check in/out time, travel intent [15], and points of interest [20]. Such information can be directly processed to generate a recommendation that optimizes some prespecified criteria or can be compared with the criteria of other travelers in a collaborative filtering method to generate hotel recommendations based on selections by travelers with similar preferences. Yadegaridehkordi et al. [21] proposed a hybrid structural equation modeling (SEM)-adaptive neuro fuzzy inference system (ANFIS) approach to find out critical factors to the success of a hotel.

Some of the relevant literature is reviewed as follows. According to the survey results of Wong and Lam [22], travelers can be divided into two clusters that exhibit different hotel selection behaviors: one cluster of leisure, short-haul, and infrequent travelers and one cluster of business, long-haul, and frequent travelers. Methods based on collaborative filtering are still in the mainstream of research in this field [23]. For example, Huming

and Weili [16] established a hotel recommendation system based on collaborative filtering. To recommend a suitable hotel to a new traveler, the rankboost algorithm [24] was applied to combine the ranking results by using data on prior travelers with similar preferences. Clustering travelers and then providing various services to the clusters, such as various transportation services, can increase travelers' satisfaction [25]. However, to achieve this, a variety of metric and nominal variables must be considered. Per Zhang et al. [15], travel intent is a major piece of additional information necessary to make a hotel recommendation. Lin et al. [26] designed an app that can record the browsing behavior of a traveler reading hotel reviews using a smartphone. Results are then analyzed using text mining techniques to determine the interests of the traveler. On the basis of this analysis, the most suitable hotel is recommended to the traveler. A similar methodology was also adopted by Boo and Busser [27]. Chen and Chuang [28] proposed an FNP approach to derive the values of weights in an FWA mechanism suitable for explaining most travelers' behaviors.

In this study, the FUTCHR system clusters travelers according to their decision-making mechanisms, which is a novel treatment in this field. Table 1 summarizes the differences between the FUTCHR system and some existing methods. From this table, it is obvious that the FUTCHR system requires the fewest inputs, considers more types of uncertainty, and is able to cluster travelers by differentiating their decision-making mechanisms based on only a single input (i.e., location).

3. Proposed methodology

3.1. Operational procedure of the FUTCHR system

The FUTCHR system is composed of four major components: a dedicated app, the central control unit, the databases (including the system database and a geodatabase), and the model builder/optimizer. The operational procedure of the FUTCHR system is illustrated in Fig. 1, and includes the following steps:

Step 1. A traveler accesses the FUTCHR system by using the dedicated app on his or her smartphone.

Step 2. The central control unit records the location of the traveler in the system database.

Step 3. The model builder/optimizer retrieves data from the system database and a geodatabase to formulate the FMBNLP model to form traveler clusters and fit their decision-making mechanisms.

Step 4. The model builder/optimizer calls the Lingo dynamic link library to solve the FMBNLP problem.

Step 5. The central control unit applies the fitted decision-making mechanisms of all clusters and recommends the hotels suitable to the traveler.

Step 6. The traveler selects a hotel and the selection is fed back into the system.

3.2. FMBNLP Model for forming traveler clusters and fitting decision-making mechanisms

The FMBNLP model separates travelers into G clusters. It is assumed that the travelers of cluster g apply the following weighted average (WA) mechanism to evaluate the overall performance of a hotel:

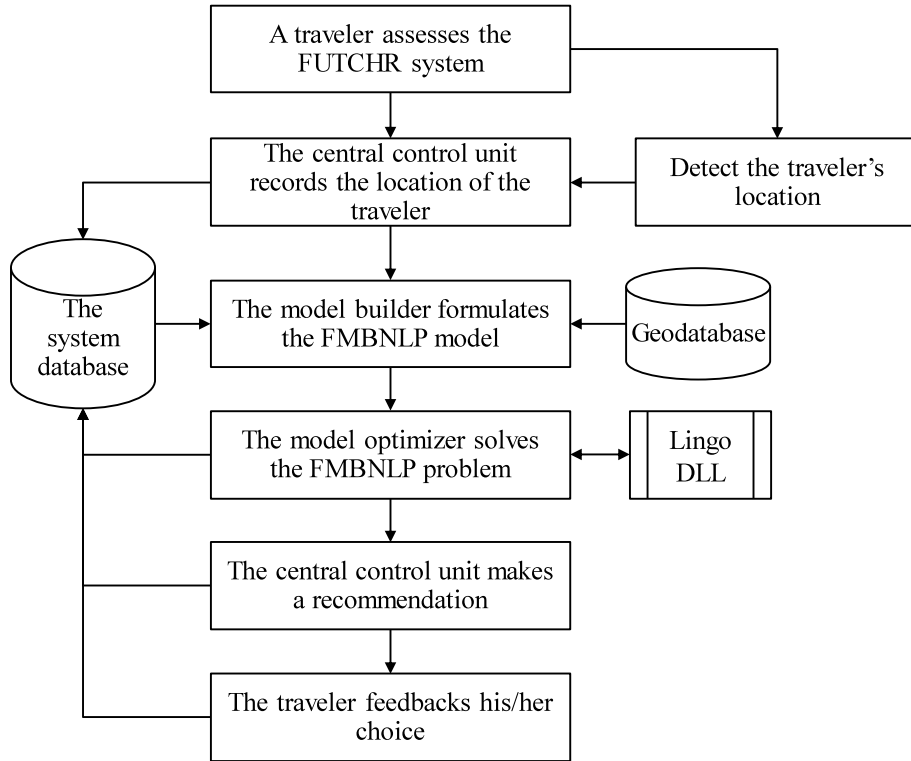
$$\tilde{\psi}_{ij}(g) = \frac{\sum_{q=1}^Q \tilde{w}_q(g) (\times) \tilde{\xi}_{ijq}}{\sum_{q=1}^Q \tilde{w}_q(g)} \quad (1)$$

where $\tilde{\psi}_{ij}(g)$ is the overall performance of hotel j that is near traveler i according to the decision-making mechanism of cluster

Table 1

Differences between the proposed methodology and some existing methods.

Method	Required inputs from traveler	Are travelers clustered?	Traveler clustering is based on	Uncertainty considered
Diana and Pronello [25]	- Location - Preferences - Demographic data	Yes	- The travel intent - Location - Mobility - Preferences - The socioeconomic condition	–
Huming and Weili [16]	- Location - Preferences	No	–	–
García-Crespo et al. [29]	- Location - Preferences - Choice	No	–	Uncertainty in choosing hotels
Levi et al. [9]	- Location - Preferences - Demographic data	Yes	- Favored reviews - Trip intent - Preferences - Nationality	–
Lin et al. [26]	- Location - The browsing behavior - Browsed reviews	No	–	–
Chen and Chuang [28]	Location	No	–	Uncertainty in choosing hotels
The FUTCHR system	Location	Yes	The decision-making mechanism	- Uncertainty in choosing hotels - Uncertainty in clustering travelers

**Fig. 1.** Operational procedure of the FUTCHR system.

$g; j = 1 \sim m_i; i = 1 \sim n; g = 1 \sim G$. $\tilde{\xi}_{ijq}$ is the q th attribute-level performance of hotel j ; $q = 1 \sim Q$. $\tilde{w}_q(g)$ is the weight assigned to attribute q ; $\tilde{w}_q(g) \in [0, 1]$. ($\times$) indicates fuzzy multiplication. However, $\tilde{w}_q(g)$ is unknown to the system administrator and must be estimated. Under the assumption that $\sum_{q=1}^Q \tilde{w}_q(g) = 1$,

$$\tilde{\psi}_{ij}(g) = \sum_{q=1}^Q \tilde{w}_q(g) (\times) \tilde{\xi}_{ijq} \quad (2)$$

If hotel α_i is chosen by traveler i ,

$$\tilde{\psi}_{i\alpha_i}(g) = \max_j \tilde{\psi}_{ij}(g) \quad (3)$$

Namely,

$$\sum_{q=1}^Q \tilde{w}_q(g) (\times) \tilde{\xi}_{i\alpha_i q} \geq \sum_{q=1}^Q \tilde{w}_q(g) (\times) \tilde{\xi}_{ijq}; \quad j \neq \alpha_i \quad (4)$$

On the server side, the system administrator guesses the values of the weights on which basis the most preferable hotel is recommended to traveler i . However, because the cluster to which traveler i belongs is unknown, the recommendation mechanisms of all clusters are applied to generate the recommendation of a suitable hotel to the traveler. As a result, G hotels, indicated with $\{\beta_i(g) | g = 1 \sim G\}$, are recommended to traveler i . The G hotels are the top-performing hotels in the corresponding traveler clusters.

The recommendation is considered successful if the hotel selected by the traveler is among the G recommended hotels:

$$I_i(g) = \begin{cases} 1 & \text{if } \alpha_i = \beta_i(g) \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

which can be converted into

$$I_i(g) \leq \frac{\sum_{j \neq \alpha_i} X_{i\alpha_{ij}}(g)}{m_i - 1} \quad (6)$$

where

$$I_i(g) \in \{0, 1\} \quad (7)$$

$$\sum_{q=1}^Q \tilde{w}_q(g)(\times) \tilde{\xi}_{i\alpha_{ij}q} \geq X_{i\alpha_{ij}}(g) \sum_{q=1}^Q \tilde{w}_q(g)(\times) \tilde{\xi}_{ijq} \quad (8)$$

$$X_{i\alpha_{ij}}(g) \in \{0, 1\} \quad (9)$$

If hotel α_i outperforms hotel j , then $X_{i\alpha_{ij}} = 1$ according to (7). Subsequently, if hotel α_i outperforms $m_i - 1$ hotels, then $I_i(g) = 1$ according to (5). For this traveler, the successful recommendation rate s_i is evaluated as

$$s_i = \frac{1}{G} \sum_{g=1}^G I_i(g) \quad (10)$$

The objective function is to optimize the average satisfaction level of all travelers:

$$\text{Max } s = \frac{1}{n} \sum_{i=1}^n s_i \quad (11)$$

Finally, the following FMBNLP model is optimized:

$$\text{Max } s = \frac{1}{n} \sum_{i=1}^n s_i \quad (12)$$

subject to

$$s_i = \frac{1}{G} \sum_{g=1}^G I_i(g); \quad i = 1 \sim n \quad (13)$$

$$I_i(g) \leq \frac{\sum_{j \neq \alpha_i} X_{i\alpha_{ij}}(g)}{m_i - 1}; \quad i = 1 \sim n; g = 1 \sim G \quad (14)$$

$$\sum_{q=1}^Q \tilde{w}_q(g)(\times) \tilde{\xi}_{i\alpha_{ij}q} \geq X_{i\alpha_{ij}}(g) \sum_{q=1}^Q \tilde{w}_q(g)(\times) \tilde{\xi}_{ijq}; \quad j \neq \alpha_i; i = 1 \sim n; g = 1 \sim G \quad (15)$$

$$\sum_{q=1}^Q \tilde{w}_q(g) = 1; g = 1 \sim G \quad (16)$$

$$0 \leq \tilde{w}_q(g) \leq 1; q = 1 \sim Q; g = 1 \sim G \quad (17)$$

$$I_i(g) \in \{0, 1\}; i = 1 \sim n; g = 1 \sim G \quad (18)$$

$$X_{i\alpha_{ij}}(g) \in \{0, 1\}; i = 1 \sim n; g = 1 \sim G \quad (19)$$

However, the FMBNLP is not directly solvable, and must be converted into a crisp form [30] as described hereafter.

When all fuzzy parameters and variables are given in triangular fuzzy numbers, constraint (14) is converted using the center of gravity (COG) method [31] to

$$\frac{\sum_{v=1}^3 \sum_{q=1}^Q (w_{qv}(g) \xi_{i\alpha_{ij}qv})}{3} \geq \frac{X_{i\alpha_{ij}}(g) \sum_{v=1}^3 \sum_{q=1}^Q (w_{qv}(g) \xi_{ijqv})}{3}; \quad j \neq \alpha_i; i = 1 \sim n; g = 1 \sim G \quad (20)$$

Eq. (15) is fuzzily satisfied by forcing the sum of the cores to 1

$$\sum_{q=1}^Q w_{q2}(g) = 1; g = 1 \sim G \quad (21)$$

Constraint (16) is expanded to

$$0 \leq w_{q1}(g) \leq 1; q = 1 \sim Q; g = 1 \sim G \quad (22)$$

$$0 \leq w_{q2}(g) \leq 1; q = 1 \sim Q; g = 1 \sim G \quad (23)$$

$$0 \leq w_{q3}(g) \leq 1; q = 1 \sim Q; g = 1 \sim G \quad (24)$$

Finally, the following MBNLP problem is to be solved instead:

$$\text{Max } s = \frac{1}{n} \sum_{i=1}^n s_i \quad (25)$$

subject to

$$s_i = \frac{1}{G} \sum_{g=1}^G I_i(g); i = 1 \sim n \quad (26)$$

$$I_i(g) \leq \frac{\sum_{j \neq \alpha_i} X_{i\alpha_{ij}}(g)}{m_i - 1}; i = 1 \sim n; g = 1 \sim G \quad (27)$$

$$\frac{\sum_{v=1}^3 \sum_{q=1}^Q (w_{qv}(g) \xi_{i\alpha_{ij}qv})}{3} \geq \frac{X_{i\alpha_{ij}}(g) \sum_{v=1}^3 \sum_{q=1}^Q (w_{qv}(g) \xi_{ijqv})}{3}; \quad j \neq \alpha_i; i = 1 \sim n; g = 1 \sim G \quad (28)$$

$$\sum_{q=1}^Q w_{q2}(g) = 1; g = 1 \sim G \quad (29)$$

$$0 \leq w_{q1}(g) \leq 1; q = 1 \sim Q; g = 1 \sim G \quad (30)$$

$$0 \leq w_{q2}(g) \leq 1; q = 1 \sim Q; g = 1 \sim G \quad (31)$$

$$0 \leq w_{q3}(g) \leq 1; q = 1 \sim Q; g = 1 \sim G \quad (32)$$

$$w_{q1}(g) \leq w_{q2}(g) \leq w_{q3}(g); q = 1 \sim Q; g = 1 \sim G \quad (33)$$

$$I_i(g) \in \{0, 1\}; i = 1 \sim n; g = 1 \sim G \quad (34)$$

$$X_{i\alpha_{ij}}(g) \in \{0, 1\}; i = 1 \sim n; g = 1 \sim G \quad (35)$$



Fig. 2. Experimental region.

Table 2
Hotels near the three travelers.

Traveler No. (<i>i</i>)	Hotel No. (<i>j</i>)
1	B, D, G, H
2	A, B, H
3	D, E, G, H

Table 3
Performances of each hotel along two attribute dimensions.

<i>J</i>	$\tilde{\xi}_{ij1}$	$\tilde{\xi}_{ij2}$
A	(0.229, 0.470, 0.715)	(0.278, 0.553, 0.588)
B	(0.243, 0.425, 0.644)	(0.201, 0.259, 0.295)
D	(0.252, 0.360, 0.593)	(0.217, 0.449, 0.516)
E	(0.081, 0.368, 0.411)	(0.159, 0.374, 0.401)
G	(0.292, 0.481, 0.752)	(0.139, 0.357, 0.590)
H	(0.180, 0.291, 0.377)	(0.138, 0.280, 0.460)

Table 4
Recommendation results.

<i>i</i>	$\beta_i(1)$	$\beta_i(2)$
1	D	G
2	A	A
3	D	G

Fig. 2), denoted as Hotels A to X, and distributed in an area of 2.3 km². Basic data concerning the hotels, namely room price, average rating, number of stars, and stylish appearance, are summarized in Table 5.

The four attributes of hotel *j* are indicated by $x_{j1} \sim x_{j4}$. The expression $x_{j3} = 0$ applies if the number of stars is NA. To convert the attributes into scores, the following formulae were applied:

$$\tilde{\xi}_{ij1} = 1 - \frac{x_{j1} - \min_l x_{l1}}{\max_l x_{l1} - \min_l x_{l1}} \quad (38)$$

$$\tilde{\xi}_{ij2} = \frac{x_{j2} - \min_l x_{l2}}{\max_l x_{l2} - \min_l x_{l2}} \quad (39)$$

$$\tilde{\xi}_{ij3} = \frac{x_{j3} - \min_l x_{l3}}{\max_l x_{l3} - \min_l x_{l3}} \quad (40)$$

$$\tilde{\xi}_{ij4} = \begin{cases} (0, 0, 0.3) & x_4 = \text{"Very Unstylish"} \\ (0, 0.3, 0.5) & x_4 = \text{"Somewhat Unstylish"} \\ (0.2, 0.5, 0.8) & \text{if } x_4 = \text{"Moderate"} \\ (0.5, 0.7, 1) & x_4 = \text{"Somewhat Stylish"} \\ (0.7, 1, 1) & x_4 = \text{"Very Stylish"} \end{cases} \quad (41)$$

Eq. (40) is based on the recommendation of Chen and Wang [32].

The dedicated app for the FUTCHR system, shown in Fig. 3, was uploaded onto app hubs such as Google Play. During the experiment, the experimental team asked for the consent of twenty travelers in the experimental area to download and use the dedicated app. The age, gender, and travel intent of each traveler have been recorded by the experimenter, as summarized in Table 6. Other characteristics such as the occupation, nationality, and preferences of a traveler have not been collected. The experimental region was close to two national universities, Hsinchu Scientific Park, and a night market, and was on the way to a popular sightseeing port. Some travelers decided to stay after visiting the night market or coming back from the sightseeing port. There were also travelers who would like to switch to a different hotel for an extended stay. The requests generated by the travelers were processed individually. The data concerning the first 10 travelers were used as the training data to construct the MBNLP model. The other travelers' data were reserved for use

3.3. A numerical example

Assume there are three travelers; $n = 3$. The hotels near the travelers are summarized in Table 2; $m_i = \{3, 2, 3\}$. The performances of each hotel along two attribute dimensions are recorded in Table 3; $Q = 2$. The hotels selected by the travelers, $\{\alpha_i\}$, are hotels D, A, and D, respectively.

Assume two traveler clusters are formed; $G = 2$. The MBNLP model for deriving the fuzzy weights is coded and solved using Lingo. The optimal solution is $\tilde{w}_1(1) = (0.028, 0.058, 0.161)$, $\tilde{w}_2(1) = (0.349, 0.942, 0.988)$, $\tilde{w}_1(2) = (0.334, 0.405, 0.464)$, and $\tilde{w}_2(2) = (0.392, 0.595, 0.743)$, giving $s^* = 0.667$. Therefore, the decision-making mechanisms adopted by the two traveler clusters are

$$\tilde{\psi}_{ij}(1) = (0.028, 0.058, 0.161)(\times)\tilde{\xi}_{ij1}(+) (0.349, 0.942, 0.988)(\times)\tilde{\xi}_{ij2} \quad (36)$$

$$\tilde{\psi}_{ij}(2) = (0.334, 0.405, 0.464)(\times)\tilde{\xi}_{ij1}(+) (0.392, 0.595, 0.743)(\times)\tilde{\xi}_{ij2} \quad (37)$$

respectively. The two decision-making mechanisms are clearly different. After applying the two decision-making mechanisms, the recommendation results issued to the travelers are summarized in Table 4.

4. Regional experiment

To assess the effectiveness of the proposed methodology, a regional experiment was conducted in Hsinchu City, Taiwan. Twenty-four hotels were located in the experimental region (see

Table 5
Basic data of the hotels.

Hotel	Room price (NTD/night)	Average rating	No. of stars	Stylish appearance
A	1152	3.7	NA	Stylish
B	1300	3.2	3	Very Unstylish
C	1100	3.6	NA	Somewhat Unstylish
D	1767	3.8	3	Very Stylish
E	1078	3.4	3	Very Unstylish
F	2206	3.8	4	Very Stylish
G	1700	4.8	NA	Moderate
H	1748	3.9	5	Stylish
I	1867	3.3	4	Moderate
J	2507	4.3	NA	Very Stylish
K	1288	3.9	3	Somewhat Unstylish
L	1984	3.7	4	Moderate
M	858	3.2	3	Somewhat Unstylish
N	1682	3.8	3	Very Unstylish
O	880	3.8	NA	Somewhat Unstylish
P	2590	4.1	4	Somewhat Unstylish
Q	748	3.5	NA	Somewhat Unstylish
R	841	3.6	NA	Somewhat Unstylish
S	850	3	2	Somewhat Unstylish
T	718	3.5	NA	Moderate
U	1801	3.8	NA	Somewhat Unstylish
V	2818	4.2	3	Moderate
W	3578	3.8	4	Very Stylish
X	3537	4.2	5	Somewhat Stylish

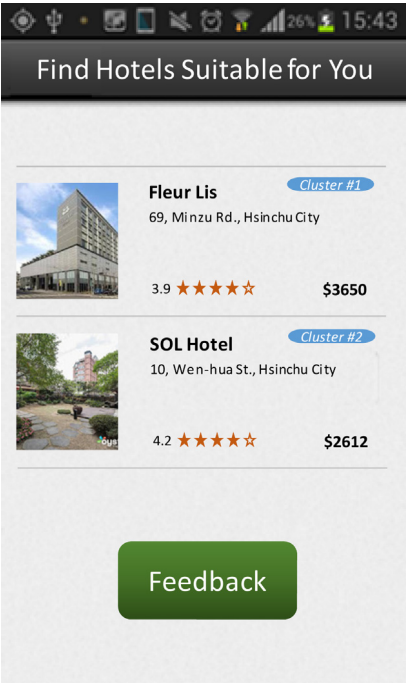


Fig. 3. The dedicated app.

as the testing data to evaluate the effectiveness of the FUTCHR system.

First, the locations of the travelers were detected using the GPS unit on their smartphones. These results are depicted in Fig. 4. Only the five hotels that were closest to a traveler according to the shortest travel time estimated by Google Maps were considered. The hotels considered by the travelers are summarized in Table 7.

At the beginning, travelers were not clustered. The FWA with equal weights was applied to derive the overall performance of a hotel:

$$\tilde{\psi}_{ij} = \frac{\sum_{q=1}^4 \tilde{\xi}_{ijq}}{4} \tag{42}$$

Table 6
Summary of the travelers' data.

Data	Results
Age	20~30: 3
	30~40: 9
	40~50: 8
Gender	Male: 11
	Female: 9
Travel Intent	Single Trip: 6
	Family/Group Trip: 11
	Business: 3

Table 7
Hotels considered by the travelers.

Traveler No.	Considered hotels
1	A, B, C, D, F
2	A, B, C, D, G
3	B, C, D, E, G
4	F, H, I, J, K
5	B, D, E, F, G
6	D, E, G, M, N
7	F, G, I, J, K
8	F, G, I, J, K
9	F, H, I, J, K
10	M, N, Q, R, S
11	K, G, L, P, S
12	N, O, R, S, T
13	I, J, K, L, P
14	M, N, Q, R, S
15	L, M, P, Q, R
16	L, P, U, W, X
17	L, M, P, Q, R
18	O, Q, R, S, T
19	L, P, U, W, X
20	L, P, U, W, X

The top performing hotel was recommended to the traveler. After receiving this, the traveler made a choice, and fed back his/her choice to the system through the dedicated app, regardless of whether he/she followed the recommendation or not. The experimenter relied on the feedback of a traveler to determine whether he/she stayed in the recommended hotel or not.

The recommended hotels to the first ten travelers, associated with their choices, are displayed in Table 8.



Fig. 4. Detected locations of the travelers.

Table 8

Hotels selected by travelers in the training data.

Traveler No.	Recommendation (β_i)	Choice (α_i)
1	F	F
2	D	D
3	D	D
4	H	H
5	F	G
6	D	M
7	F	F
8	F	G
9	H	H
10	M	M

Table 9

Hotels recommended to travelers in the testing data.

Traveler No.	Recommended hotels	Choice (α_i)	Cluster
11	L, K	K	2
12	N, N	N	1 or 2
13	L, J	L	1
14	M, M	M	1 or 2
15	M, M	L	–
16	L, L	L	1 or 2
17	M, M	Q	–
18	Q, Q	S	–
19	L, L	L	1 or 2
20	L, L	L	1 or 2

Table 10

Recommendation results using the nearest hotel method (the top k policy).

Traveler No.	Recommended hotels	Choice (α_i)
11	L, K	K
12	O, T	N
13	J, K	L
14	Q, R	M
15	P, Q	L
16	U, W	L
17	P, Q	Q
18	T, S	S
19	W, U	L
20	X, W	L

The data of the first ten travelers were used to construct the MBNLP model. Two traveler clusters were formed; $G = 2$. Lingo 16.0 $\times$ 64 was employed to construct the MBNLP problem and implement a branch-and-bound algorithm to solve the MBNLP problem on a PC with Intel Core i7-7700 CPU 3.60 GHz and 8 GB RAM within 1 min. The optimal values of weights for the two traveler clusters were derived as

$$\tilde{w}_1^*(1) = (0.162, 0.233, 0.599)$$

$$\tilde{w}_2^*(1) = (0.082, 0.154, 0.294)$$

$$\tilde{w}_3^*(1) = (0.233, 0.552, 0.651)$$

$$\tilde{w}_4^*(1) = (0.046, 0.061, 0.078)$$

$$\tilde{w}_1^*(2) = (0.024, 0.109, 0.191)$$

$$\tilde{w}_2^*(2) = (0.059, 0.112, 0.443)$$

$$\tilde{w}_3^*(2) = (0.104, 0.288, 0.441)$$

$$\tilde{w}_4^*(2) = (0.168, 0.491, 0.501)$$

yielding an optimal objective function value of 0.75. For the two traveler clusters, the priorities concerning the attributes of a hotel varied considerably:

Cluster 1: No. of Stars > Room Price > Average Rating > Stylish Appearance

Cluster 2: Stylish Appearance > No. of Stars > Average Rating > Room Price

Subsequently, the decision-making mechanisms of the two traveler clusters were applied to the testing data. The recommendation results are summarized in Table 8. After considering the recommendations, the travelers made their choices, which are also presented in Table 9. The successful recommendation

rate was 60%. By comparing the recommended hotel to the actual choice, the cluster that a traveler belonged to could be ascertained. These results are also depicted in Table 8.

For comparison, the collected data (including the location and choice of each traveler) were also analyzed using the recommendation mechanisms of three existing methods. The first method was the nearest hotel method that recommended the nearest hotel to a traveler, similar to the recommendation mechanism adopted by Google Maps. The proposed methodology recommended two hotels to a traveler. To achieve a fair comparison, in the nearest hotel method the two nearest hotels were both recommended to a traveler; the top k policy. The resulting recommendations received by travelers in the testing data are summarized in Table 10. The successful recommendation rate was 10%.

The second method was the ordered weighted average (OWA) method [33]. In the OWA method, the weights of attributes were determined after, not before, observing the performances. Then, a criterion exhibiting superior performance was assigned a higher weight. In the regional study, the “moderately optimistic”

Table 11
Recommendation results using the ordered weighted average (OWA) method (the top k policy).

Traveler No.	Recommended hotels	Choice (α_i)
11	G, K	K
12	N, R	N
13	J, K	L
14	N, R	M
15	R, P	L
16	P, U	L
17	R, P	Q
18	R, Q	S
19	P, U	L
20	P, U	L

Table 12
Recommendation results using the FNP method (the top k policy).

Traveler No.	Recommended hotels	Choice (α_i)
11	L, K	K
12	N, O	N
13	I, L	L
14	M, Q	M
15	M, Q	L
16	L, P	L
17	M, Q	Q
18	Q, O	S
19	L, P	L
20	L, P	L

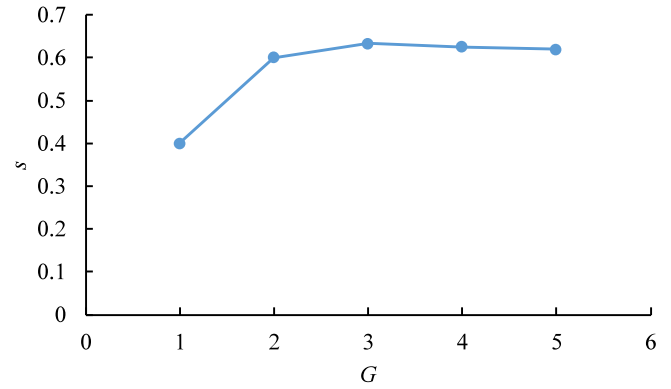


Fig. 5. The parametric analysis results.

decision strategy was adopted. The weights assigned to the top four criteria were 0.660, 0.153, 0.105, and 0.083, respectively. A fuzzy performance was defuzzified using the COG method before sorting the performances and applying the weights. The top two performing hotels were recommended to each traveler. The recommendation results using the OWA method are summarized in Table 11. The successful recommendation rate was 5%.

The third method was the FNP approach proposed by Chen and Chuang [28]. The FNP approach derived the fuzzy weights by maximizing the sum of the differences between the selected hotel and other hotels. Travelers were not divided into clusters. After optimization, the following decision-making mechanism was fitted:

$$\tilde{\psi}_{ij} = (0.048, 0.082, 0.271)(\times)\tilde{\xi}_{ij1}(+)(0.104, 0.137, 0.289)(\times)\tilde{\xi}_{ij2} \\ (+)(0.210, 0.421, 0.527)(\times)\tilde{\xi}_{ij3}(+)(0.239, 0.360, 0.510)(\times)\tilde{\xi}_{ij4} \quad (43)$$

This yielded an optimal objective function value of 22.67 for the training data. Subsequently, the fitted decision-making mechanism was applied to the testing data. For a fair comparison, the top k policy was followed in implementing the FNP method, and the top two performing hotels were both recommended to a traveler. The recommendation results are summarized in Table 12. The successful recommendation rate was 40%.

According to the experimental results, three primary conclusions can be drawn:

- (1) Among the four compared methods, the proposed methodology achieved the highest successful recommendation rate for travelers in the testing data, which supported the effectiveness of the FUTCHR system.
- (2) The application of the OWA method yielded the lowest successful recommendation rate, which implied that travelers might focus attention on hotel attributes with poor performances rather than those with superior performances.
- (3) The effectiveness of clustering travelers was evident in the superiority of the proposed methodology (with traveler

clustering) over the FNP approach (without traveler clustering), achieving a 20% higher successful recommendation rate. This indicated that all travelers did not adopt the same the decision-making mechanisms. As a result, the performance of a recommendation mechanism that did not cluster travelers, such as the FNP method, worsened as the number of travelers increases (from ten to twenty). In contrast, the FUTCHR system clustered travelers to maintain a satisfactory performance as the number of travelers increased, thereby tackling the scalability problem.

- (4) In Table 8, among travelers in the testing data, travelers #11 and #13 behaved different from other travelers. The proposed methodology succeeded in recommending suitable hotels to the two travelers, while the nearest hotel method and the OWA method failed, which showed that the proposed methodology was able to solve the cold-start (or sparsity) problem.
- (5) The number of traveler clusters (G) has been varied to observe the performance of the proposed methodology. The recommendation mechanisms with various values of G are summarized in Table 13. As more traveler clusters were formed, more hotels were recommended to a traveler, thereby increasing the possibility that the traveler chose from the recommended hotels. However, that did not necessarily elevate the successful recommendation rate since the successful recommendation rate was inversely proportional to the number of recommended hotels (i.e., G). The parametric analysis results are shown in Fig. 5. When three traveler clusters were formed, the proposed methodology achieved the highest satisfaction rate for the testing data.
- (6) The proposed methodology (i.e., the three recommendation mechanisms fitted using the data of travelers #1 ~ #10) has been applied to twenty new travelers to further elaborate its effectiveness. The recommended hotels and choices of the new travelers are summarized in Table 14. The successful recommendation rate was only 40%. The reason might be the changes in the decision-making mechanisms of the new travelers. To address this issue, the recommendation mechanisms were updated by mining the data of travelers #11 ~ #20. Then, the updated recommendation mechanisms were applied to the data of the new travelers. The recommendation results are presented in Table 15. The successful recommendation rate was elevated to 55%.

5. Conclusions

This study investigated a topic of major importance, ubiquitous hotel recommendation, and determined the decision-making

Table 13

The recommendation mechanisms with various numbers of clusters.

G	The recommendation mechanisms
2	$\tilde{\psi}_{ij}(1) = (0.162, 0.233, 0.599)(\times)\tilde{\xi}_{ij1}(+) (0.082, 0.154, 0.294)(\times)\tilde{\xi}_{ij2}$ $(+) (0.233, 0.552, 0.651)(\times)\tilde{\xi}_{ij3}(+) (0.046, 0.061, 0.078)(\times)\tilde{\xi}_{ij4}$
	$\tilde{\psi}_{ij}(2) = (0.024, 0.109, 0.191)(\times)\tilde{\xi}_{ij1}(+) (0.059, 0.112, 0.443)(\times)\tilde{\xi}_{ij2}$ $(+) (0.104, 0.288, 0.441)(\times)\tilde{\xi}_{ij3}(+) (0.168, 0.491, 0.501)(\times)\tilde{\xi}_{ij4}$
3	$\tilde{\psi}_{ij}(1) = (0.148, 0.303, 0.304)(\times)\tilde{\xi}_{ij1}(+) (0.147, 0.201, 0.281)(\times)\tilde{\xi}_{ij2}$ $(+) (0.142, 0.227, 0.293)(\times)\tilde{\xi}_{ij3}(+) (0.113, 0.269, 0.288)(\times)\tilde{\xi}_{ij4}$
	$\tilde{\psi}_{ij}(2) = (0.111, 0.317, 0.453)(\times)\tilde{\xi}_{ij1}(+) (0.136, 0.319, 0.433)(\times)\tilde{\xi}_{ij2}$ $(+) (0.223, 0.312, 0.322)(\times)\tilde{\xi}_{ij3}(+) (0.043, 0.052, 0.367)(\times)\tilde{\xi}_{ij4}$
	$\tilde{\psi}_{ij}(3) = (0.067, 0.335, 0.345)(\times)\tilde{\xi}_{ij1}(+) (0.131, 0.208, 0.211)(\times)\tilde{\xi}_{ij2}$ $(+) (0.066, 0.338, 0.374)(\times)\tilde{\xi}_{ij3}(+) (0.043, 0.118, 0.356)(\times)\tilde{\xi}_{ij4}$
	$\tilde{\psi}_{ij}(4) = (0.125, 0.197, 0.229)(\times)\tilde{\xi}_{ij1}(+) (0.014, 0.282, 0.312)(\times)\tilde{\xi}_{ij2}$ $(+) (0.121, 0.287, 0.331)(\times)\tilde{\xi}_{ij3}(+) (0.136, 0.234, 0.298)(\times)\tilde{\xi}_{ij4}$
4	$\tilde{\psi}_{ij}(2) = (0.099, 0.216, 0.280)(\times)\tilde{\xi}_{ij1}(+) (0.164, 0.332, 0.368)(\times)\tilde{\xi}_{ij2}$ $(+) (0.113, 0.211, 0.302)(\times)\tilde{\xi}_{ij3}(+) (0.005, 0.241, 0.273)(\times)\tilde{\xi}_{ij4}$
	$\tilde{\psi}_{ij}(3) = (0.181, 0.242, 0.340)(\times)\tilde{\xi}_{ij1}(+) (0.008, 0.259, 0.358)(\times)\tilde{\xi}_{ij2}$ $(+) (0.172, 0.228, 0.296)(\times)\tilde{\xi}_{ij3}(+) (0.013, 0.272, 0.389)(\times)\tilde{\xi}_{ij4}$
	$\tilde{\psi}_{ij}(4) = (0.048, 0.172, 0.318)(\times)\tilde{\xi}_{ij1}(+) (0.070, 0.230, 0.398)(\times)\tilde{\xi}_{ij2}$ $(+) (0.138, 0.347, 0.392)(\times)\tilde{\xi}_{ij3}(+) (0.199, 0.251, 0.373)(\times)\tilde{\xi}_{ij4}$
	$\tilde{\psi}_{ij}(1) = (0.028, 0.291, 0.304)(\times)\tilde{\xi}_{ij1}(+) (0.126, 0.131, 0.317)(\times)\tilde{\xi}_{ij2}$ $(+) (0.077, 0.285, 0.343)(\times)\tilde{\xi}_{ij3}(+) (0.006, 0.293, 0.309)(\times)\tilde{\xi}_{ij4}$
	$\tilde{\psi}_{ij}(2) = (0.163, 0.194, 0.215)(\times)\tilde{\xi}_{ij1}(+) (0.099, 0.282, 0.320)(\times)\tilde{\xi}_{ij2}$ $(+) (0.038, 0.217, 0.316)(\times)\tilde{\xi}_{ij3}(+) (0.148, 0.306, 0.309)(\times)\tilde{\xi}_{ij4}$
5	$\tilde{\psi}_{ij}(3) = (0.026, 0.063, 0.244)(\times)\tilde{\xi}_{ij1}(+) (0.092, 0.226, 0.392)(\times)\tilde{\xi}_{ij2}$ $(+) (0.219, 0.275, 0.462)(\times)\tilde{\xi}_{ij3}(+) (0.245, 0.437, 0.443)(\times)\tilde{\xi}_{ij4}$
	$\tilde{\psi}_{ij}(4) = (0.032, 0.104, 0.130)(\times)\tilde{\xi}_{ij1}(+) (0.023, 0.351, 0.367)(\times)\tilde{\xi}_{ij2}$ $(+) (0.177, 0.362, 0.365)(\times)\tilde{\xi}_{ij3}(+) (0.167, 0.183, 0.260)(\times)\tilde{\xi}_{ij4}$
	$\tilde{\psi}_{ij}(5) = (0.118, 0.139, 0.463)(\times)\tilde{\xi}_{ij1}(+) (0.036, 0.207, 0.498)(\times)\tilde{\xi}_{ij2}$ $(+) (0.017, 0.118, 0.312)(\times)\tilde{\xi}_{ij3}(+) (0.117, 0.536, 0.542)(\times)\tilde{\xi}_{ij4}$

Table 14

The recommended hotels and choices of the travelers.

Traveler #	Recommended hotels	Choices
21	{N, N, N}	O
22	{F, F, F}	F
23	{M, L, M}	Q
24	{H, H, H}	F
25	{H, H, H}	H
26	{N, N, N}	O
27	{L, L, L}	L
28	{F, F, F}	I
29	{M, L, M}	M
30	{M, L, M}	M
31	{F, F, F}	J
32	{L, L, L}	P
33	{N, N, N}	O
34	{Q, Q, Q}	O
35	{Q, Q, Q}	Q
36	{N, N, N}	T
37	{L, L, L}	G
38	{H, H, H}	H
39	{M, L, M}	M
40	{Q, Q, Q}	Q

mechanisms of travelers may vary considerably. Therefore, generating recommendations that conform to the decision-making mechanisms of all travelers, especially when only a single recommendation mechanism is adopted on the server side, presented a highly challenging task. To overcome this problem, the FUTCHR system was developed in this study. The FUTCHR system detected the location of a traveler and did not need additional inputs from the traveler. The FUTCHR system divided travelers into multiple clusters before recommending them suitable hotels. To travelers in a cluster, the same recommendation mechanism was applied. However, the recommendation mechanisms for various clusters were not equal. In the proposed methodology, the recommendation mechanisms for the traveler clusters were in the form of FWA but had various fuzzy weights. To determine the fuzzy

Table 15

The recommendation results after updating the recommendation mechanisms.

Traveler #	Recommended hotels
21	{N, O, O}
22	{F, F, F}
23	{M, M, M}
24	{H, H, H}
25	{H, H, H}
26	{N, O, O}
27	{L, G, G}
28	{F, F, F}
29	{M, M, M}
30	{M, M, M}
31	{F, F, F}
32	{L, L, L}
33	{N, O, O}
34	{Q, Q, Q}
35	{Q, Q, Q}
36	{N, O, O}
37	{L, G, G}
38	{H, H, H}
39	{M, M, M}
40	{Q, Q, Q}

weights, an FMBNLP model was proposed and converted into an MBNLP problem to be solved. The novelty of the proposed methodology is to cluster travelers without knowing their characteristics but according to the differences in their decision-making mechanisms.

A numerical example was used to illustrate the applicability of the proposed methodology. In addition, a regional study was conducted to assess the effectiveness of the FUTCHR system. According to the experimental results, three key points were discerned:

- (1) Compared with three existing methods, the FUTCHR system achieved a higher successful recommendation rate.
- (2) Travelers were divided into two clusters, which raised the successful recommendation rate by 20%.
- (3) Travelers did not focus their attention on hotel attributes with superior performances, which made OWA ineffective.
- (4) Travelers have various characteristics and preferences. Therefore, the hotels chosen by different travelers in similar locations may not be the same. To address this issue, the proposed methodology recommends multiple hotels, rather than a single hotel, to a traveler. In the existing methods, if multiple hotels are recommended, the hotels are usually the top k performing hotels. In contrast, in the proposed methodology, all the recommended hotels are the top 1 performing hotels (for k traveler clusters). As a result, travelers in similar locations may choose different hotels, depending on the clusters they belong to.
- (5) When only the location of a traveler is known, the proposed methodology is able to generate a better recommendation than the existing methods. If the characteristics and preferences of a traveler are also provided, the proposed methodology can be enhanced by taking such information into account, and the recommendation result can be better.

The effectiveness of the FUTCHR system must be tested under more environments, for which the configuration of the FUTCHR system may vary, so as to study how the proposed methodology behaves under such environments. In addition, the simple FWA recommendation mechanisms may be replaced with more sophisticated, perhaps nonlinear, recommendation mechanisms in future studies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This study was sponsored by the Ministry of Science and Technology, Taiwan.

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